

3(a)	any three from: 1 seed / embryo, absorbs water ; 2 (stimulates) embryo to produce gibberellin ; 3 gibberellin moves to aleurone layer ; 4 gene coding for amylase, expressed / transcribed ; 5 translation of mRNA (to produce amylase) ; 6 AVP ; e.g. gibberellin binds to receptor / ref. destruction of DELLA protein	3
3(b)(i)	<u>endosperm</u> ;	1

Question	Answer	Marks																					
3(b)(ii)	any four from: 1 amylase activity increases ; 2 as (seeds) produce more amylase ; 3 percentage of starch decreases ; 4 amylase, hydrolyses / breaks down, starch ; 5 <i>idea of</i> a change at an increasing rate (for either) ; A exponential change 6 data quote: amylase or starch at two different days ; <table border="1"> <thead> <tr> <th>time / days</th><th>amylase activity / au</th><th>percentage of starch remaining</th></tr> </thead> <tbody> <tr> <td>0</td><td>0.0</td><td>100</td></tr> <tr> <td>1</td><td>0.2</td><td>99</td></tr> <tr> <td>2</td><td>0.8</td><td>98</td></tr> <tr> <td>3</td><td>2.4</td><td>95</td></tr> <tr> <td>4</td><td>5.0</td><td>91</td></tr> <tr> <td>5</td><td>8.4</td><td>82</td></tr> </tbody> </table> <i>accept manipulated data</i>	time / days	amylase activity / au	percentage of starch remaining	0	0.0	100	1	0.2	99	2	0.8	98	3	2.4	95	4	5.0	91	5	8.4	82	4
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Question	Answer	Marks
3(c)(i)	any three from: 1 enzyme, denatured / deactivated / inactivated ; 2 changes, tertiary structure / 3D shape, of amylase ; A enzyme 3 changes shape of active site ; 4 active site, no longer complementary / does not bind, to starch / substrate or no, ESC / enzyme-substrate complexes, formed ; 5 AVP ; e.g. removes water from embryo for gibberellin synthesis / removes water from endosperm for hydrolytic reactions	3
3(c)(ii)	any one from: 1 stops, sugars / glucose, from being used ; ora 2 ensures (enough) sugars for beer production ;	1